Ayush Madhav Kumar

Ann Arbor, MI | +1 734-929-7412 | contactayushmadhav@gmail.com | ayushmadhav.com | linkedin.com/in/ayushmk | github.com/ayushmk7

Education

University of Michigan, College of Engineering **Aug 2025 – Dec 2028**

B.S.E. Computer Science, Minor in Mathematics; GPA 3.60/4.00 *Ann Arbor, MI*

- **Coursework:** Data Structures and Algorithms, Computer Organization, Discrete Mathematics, Linear Algebra

Experience

Jaseci Labs **May 2026 – Aug 2026**

Software Engineering Intern *Ann Arbor, MI*

- Fine-tuned Qwen3-Coder 30B into a coding agent for the Jac language, reaching 94% compile-pass on a held-out benchmark, using Unsloth and 4-bit QLoRA
- Generated 300k+ training examples through a synthetic data pipeline, each gated on a compiler and unit tests
- Cut the model's fallback to Python patterns 82% with a four-stage SFT, debugging, DPO, and agentic curriculum

NSF Pathways to Open Source Ecosystems **May 2026 – Present**

Undergraduate Researcher *Ann Arbor, MI*

- Researching a compiler and runtime path that runs Jac's graph-native walker programs on NVIDIA GPUs, lowering abilities through LLVM IR and NVPTX to PTX with one walker per thread and SoA and CSR graph packing
- Isolating layout from compiler effects in fixed-PTX, Nsight-timed experiments: access-order placement gave a 1.75× kernel speedup on 256^3 matmul, and generated kernels matched checked handwritten CUDA on 17 subtree queries
- Built Marketing Agent Studio (55 Jac modules) and NightShift, a nightly compiler-audit agent that opens PRs

Cactus (YC S25) **Jan 2026 – Present**

Open Source Contributor *Remote*

- Fixed image completions hanging on Metal and CPU: a shape bug split a 1,043-token prompt into 1,000 chunks
- Fixed two C++ tokenizer bugs: byte-fallback pieces decoded as literal text, and vocab lines mis-split on tabs
- Documented custom vocabulary and boost options for transcription across four SDKs: Python, Swift, Kotlin, Flutter

CareTether (Patch Summer Accelerator '25) **Jul 2025 – Aug 2025**

Co-Founder *Dublin, Ireland*

- Built the app front end for a senior-care wearable and ran a pilot with 5+ families over 8 weeks
- Reached 20,000+ people through social campaigns and pitched NGOs and care homes on distribution

Projects

NASA SUITS Challenge | *AI Lead, team of 6; Python, PyTorch, Unity*

- Led the AI subteam on autonomous lunar EVA navigation to 96.8% waypoint accuracy, with A*/JPS planning over a 3-layer occupancy grid and 17-ray LiDAR fusion
- Trained the obstacle detection models served in real time behind a hot-swappable backend: a temporal ResidualBeamCNN and a BEV cross-attention transformer
- Built the voice pipeline at 200ms end to end, streaming 16 kHz PCM over WebSocket into faster-whisper

OSINT Geospatial Intelligence Platform | *TypeScript, React, CesiumJS, Node.js, WebSocket*

- Onboarded any public data feed as a single YAML file with no code changes, with JSON, GeoJSON, XML and CSV parsers
- Eliminated a 372K-row duplication bug across 31K entities and 780K+ observations with a deterministic observation ID and a unique natural-key index
- Kept tens of thousands of live markers interactive on integrated GPUs, batching into one CesiumJS BillboardCollection and streaming delta-only updates

Technical Skills

- **Languages:** Python, C++, C, Java, TypeScript, JavaScript, Rust, SQL, Jac
- **Compilers and GPU:** LLVM IR, NVPTX, PTX, CUDA, Nsight Systems, SoA and CSR graph layouts
- **Machine Learning:** PyTorch, TensorFlow, HuggingFace Transformers, Unsloth, LoRA/QLoRA, DPO, vLLM, ONNX Runtime, scikit-learn, RAG, quantization
- **Backend and Infrastructure:** FastAPI, Flask, Django, Node.js, Express, REST, gRPC, GraphQL, WebSockets, RabbitMQ, Docker, Kubernetes, Linux, Amazon Web Services, Terraform, GitHub Actions
- **Data and Frontend:** PostgreSQL, MySQL, MongoDB, Redis, SQLite, Pandas, NumPy, React, Next.js, React Native, Redux Toolkit, CesiumJS, Vite, Flutter